

FINAL EXAM

ECON498–ADVANCED TOPICS IN ECONOMICS: GENERAL EQUILIBRIUM

MAY 1, 2026

1. Fix an exchange economy $\mathcal{E} = (X_i, \omega_i, \succsim_i)_{i \in N}$ and a Pareto efficient allocation x^* such that for every i ,
 - $X_i = \mathbb{R}_+^L$,
 - $\succsim_i$ is represented by a concave, continuous, monotone function $u_i : \mathbb{R}_+^L \rightarrow \mathbb{R}$ satisfying $u_i(0) = 0$,
 - $x_i^* > 0$.

Show that x^* solves the weighted welfare maximization problem

$$\begin{aligned} & \max_{(x_1, \dots, x_n)} \sum_{i \in N} \alpha_i u_i(x_i) \\ & \text{subject to } x_i \geq 0 \text{ for every } i \text{ and } \sum_{i \in N} x_i \leq \sum_{i \in N} \omega_i \end{aligned}$$

for suitably chosen weights $(\alpha_i)_{i \in N}$ satisfying $\alpha_i > 0$ for every i and

$$\sum_{i \in N} \alpha_i = 1.$$

2. Let $X \subseteq \mathbb{R}^m$ and $Y \subseteq \mathbb{R}^n$. Let $\varphi : X \rightrightarrows Y$ be a possibly empty-valued correspondence. We will say that φ satisfies the local intersection property if for every $x \in X$ with $\varphi(x) \neq \emptyset$, x has an open set V in X such that $\bigcap_{x' \in V} \varphi(x') \neq \emptyset$. Prove the following statement.

Proposition. *Let $\succ$ be an irreflexive binary relation—indicating strict preference—on a nonempty, compact and convex set $X \subseteq \mathbb{R}^m$. Define $\varphi : X \rightrightarrows X$, as usual, by $\varphi(x) = \{x' \in X : x' \succ x\}$. Suppose that $\succ$ is convex and that φ satisfies the local intersection property. Then the set of $\succ$ -maximal elements in X is nonempty.*

3. In this problem, we will verify some auxiliary results and missing steps in the proof of existence of Free Disposal Equilibrium prices with single-valued demands.

Recall that we showed the set $\{p \in \Delta : z(p) \leq 0\}$ is nonempty and compact, and we said that such prices are Free Disposal Walrasian Equilibrium (FDWE) prices.

Recall the following definition. A binary relation $\succsim$ is locally non-satiated (LNS) if for every $x \in X$ and every $\varepsilon > 0$, there is $y \in X$ such that $\|y - x\| < \varepsilon$ and $y \succ x$.

- (a) Suppose that agent i has LNS preferences. Assume the demand correspondence is single-valued, i.e., a demand function, $d_i(p)$. Assume the endowment of agent i satisfies $\omega_i > 0$. Show that if $p \in \Delta_0$ then $p \cdot d_i(p) = p \cdot \omega_i$, i.e., agent i exhausts her budget.
- (b) With two goods and two agents, suppose that $p^* \in \{p \in \Delta : z(p) \leq 0\}$ and suppose that both agents have LNS preferences. Suppose toward a contradiction that $z_1(p^*) < 0$ and $p_1 > 0$.
 - i. Show that at least one agent must be supplying good 1.
 - ii. Suppose without loss of generality that agent 1 supplies good 1, i.e., $d_{11}(p^*) < \omega_{11}$. Using (a), show that agent 1 must be demanding good 2, i.e., $d_{12}(p^*) > \omega_{12}$, and that $p_2 > 0$.
 - iii. Show that agent 2 must be supplying good 2, i.e., $d_{22}(p^*) < \omega_{22}$.
 - iv. Using (a), show that agent 2 must be demanding good 1, i.e., $d_{21}(p^*) > \omega_{21}$.
 - v. Using the budget equalities of both agents, derive a contradiction.
 - vi. Conclude that if a good is in excess supply and agents have LNS preferences, the price of such good must be zero.

4. Let $\omega_1 = (1, 2)$ and $\omega_2 = (2, 3)$. For each of the following economies:

- (a) $u_1(x, y) = x + y$, $u_2(x, y) = 2x + 3y$
- (b) $u_1(x, y) = x^{1/2}y^{1/2}$, $u_2(x, y) = x^{2/3}y^{1/3}$
- (c) $u_1(x, y) = \min\{x, y\}$, $u_2(x, y) = \min\{2x, 3y\}$
- (d) $u_1(x, y) = x + y$, $u_2(x, y) = \min\{2x, 3y\}$

(e) $u_1(x, y) = x^{1/2}y^{1/2}$, $u_2(x, y) = \min\{2x, 3y\}$

Do the following:

- Compute, for each agent i , the demand correspondence $D_i(p)$, and the excess demand correspondence $z_i(p)$. Finally, compute the excess market demand correspondence $z(p)$. Note that some of the previous objects can actually be functions.
- Find (compute and picture) the Pareto set and the core.
- Find the Walrasian Equilibrium prices and allocations. If a Walrasian Equilibrium does not exist, find a Free Disposal Walrasian Equilibrium.

Note: You may derive demands informally using the properties of each utility function (e.g., tangency conditions, corner solutions, fixed proportions) or a graphical approach. You don't need to be excessively formal here, e.g., I don't want to see Lagrangians or Karush-Kuhn-Tucker conditions.

5. Consider a consumer whose preference relation over the consumption space $X = \mathbb{R}_+^2$ is as follows:

$$x \succsim y \quad \text{if either}$$

- $\sqrt{x_1} + x_2 > \sqrt{y_1} + y_2$, or
- $\sqrt{x_1} + x_2 = \sqrt{y_1} + y_2$ and $x_1 \geq y_1$.

- For an arbitrary consumption bundle $x \in \mathbb{R}_{++}^2$, illustrate the sets $U(x)$ and $U^{-1}(x)$.
- Is the preference relation reflexive? Total? Transitive? Convex? Continuous?
- Is it representable by a utility function?
- Irrespective of your answer to the last item, recall that we can define the demand correspondence as

$$D(p, w) = \{x \in \mathbb{R}_{++}^2 : U(x) \cap \beta(p, w) = \emptyset\}$$

where $\beta : \Delta_0 \times \mathbb{R}_{++} \Rightarrow \mathbb{R}_+^2$ is the budget correspondence. Compute the demand correspondence for the binary relation above.

Note: In here, be formal, but not excessively so. Construct your arguments carefully and appeal to results we've seen where needed.

6. Consider the following demand correspondence defined over Δ_0 and $w > 0$:

$$D_1(p, w) = D_2(p, w) = \begin{cases} \frac{w}{p_1 + p_2}, & \text{if } w \leq p_1 + p_2 \\ 1, & \text{if } w > p_1 + p_2 \end{cases}$$

- Does this demand satisfy $p \cdot x(p, w) = w$ for all (p, w) ? If not, for which prices and wealth levels does the equality hold?
- Can this demand come from maximizing a utility function over a budget set? If so, find the utility function.
- Consider a two-agent exchange economy in which agent 1 has the demand correspondence above and agent 2 has preferences represented by $u_2(x_1, x_2) = x_1^{1/2}x_2^{1/2}$. Let $(\omega_1, \omega_2) \in \mathbb{R}_{++}^2 \times \mathbb{R}_{++}^2$ denote the endowments of agents 1 and 2, respectively.
 - Does a Free Disposal Walrasian Equilibrium always exist? How about a Walrasian Equilibrium? Justify your answer.
 - Characterize the set of equilibrium prices (WE or FDWE, as applicable) as a function of the endowments (ω_1, ω_2) . For which endowments is this set a singleton? For which endowments is it larger?